Divya Sreelekha Kara

AI & ML ENGINEER | COMPUTER VISION | DATA ANALYTICS | FULL-STACK AI APPLICATIONS

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PROFESSIONAL SUMMARY

Highly motivated and skilled **AI & ML Engineer** with hands-on experience in computer vision, deep learning, and full-stack AI application development. Proficient in **Python** and frameworks such as **PyTorch, TensorFlow, OpenCV, and FastAPI** to build and deploy production-grade intelligent systems. Solid understanding of **Data Structures & Algorithms (DSA)**, Object-Oriented Programming (OOP), and database management. Proven ability to build end-to-end pipelines, fine-tune large reasoning models, and deliver full-stack solutions. Seeking to leverage technical expertise in artificial intelligence, model optimization, and software engineering to drive innovative digital solutions at TCS.

TECHNICAL SKILLS

Programming Languages	Python, JavaScript, C, SQL
ML / AI & Algorithms	Computer Vision, CNNs, Deep Learning, Transfer Learning, LoRA Fine-Tuning, cGAN, Diffusion Models, Supervised & Unsupervised Learning, NLP basics
Frameworks & Libraries	PyTorch, TensorFlow, OpenCV, MediaPipe, FastAPI, Flask, NumPy, Pandas, Scikit-learn, Unsloth, TRL
Web & Databases	React.js, Node.js, Express.js, HTML5, CSS3, MongoDB, MySQL
Tools & Practices	Git, GitHub, Jupyter Notebooks, Kaggle, REST APIs, Agile basics, OWASP Awareness, Secure Coding Principles

EDUCATION

Vignan's Lara Institute of Technology & Science
2023 – 2027

B.Tech in Computer Science and Engineering (AI & ML)
CGPA: 8.7 / 10.0

- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Computer Vision, Database Management Systems, Operating Systems, OOP with Python & C++.
- Active participant in campus AI/ML hackathons, coding competitions, and technical seminars; served as team lead on multiple cross-functional project teams.

TECHNICAL PROJECTS

C-THRU — Generative AI-Based Cloud Removal & Surface Reconstruction
2026 – Present

Python, PyTorch, cGAN, Streamlit, Rasterio, Linux

- Developing an operational deep learning framework (cGAN & DDPM Diffusion) to remove persistent cloud cover and shadow contamination from high-resolution **LISS-IV optical imagery**.
- Integrating cloud-penetrating **Sentinel-1 SAR** satellite imagery for topographical feature extraction, preserving terrain details under clouds.
- Built an interactive Streamlit dashboard for resample operations, coordinate alignment, and reconstructed GeoTIFF export (evaluating PSNR, SSIM, SAM).

RadReport — AI Chest X-ray Report Generator
2026

Python, FastAPI, OpenCV, React.js

- Built a clinical vision-language model (VLM) platform with a **FastAPI** backend that auto-generates structured radiology reports from uploaded chest X-rays, reducing manual reporting time.
- Implemented a dual-mode AI engine utilizing VLM inference with rule-based clinical heuristics fallback and **OpenCV** attention heatmap overlays for explainability.
- Delivered a full-stack deployment with a **React.js** frontend, enabling clinician-friendly report review, annotation, and PDF export workflows.

Nemotron Solver — Agentic Reasoning Model Fine-Tuning
2026

Python, PyTorch, LoRA, Jupyter

- Designed data augmentation pipelines and training corpus generation scripts; applied **LoRA** adapters for parameter-efficient supervised fine-tuning (SFT) on the Nemotron reasoning model (GSM8K-style datasets).

- Built an interactive training dashboard visualizing token-level reasoning traces, loss curves, and per-problem solve rates across model checkpoints to enable rapid iteration and debugging.

HANDY — Real-Time Hand Gesture Recognition System 2025

Python, OpenCV, MediaPipe

- Engineered a real-time hand gesture recognition system achieving **99% classification accuracy** at 20+ FPS by tracking 21 hand landmarks per frame using **MediaPipe** and **OpenCV**.
- Integrated gesture-to-command mapping for interactive control interfaces, demonstrating efficient on-device AI inference on standard webcam hardware.

WORK EXPERIENCE / INTERNSHIP

Machine Learning Intern Mar 2026 – Apr 2026

SkillCraft Technology

- Implemented end-to-end supervised learning models (Linear Regression, SVM) and unsupervised models (K-Means Clustering) using **Python**, **Scikit-learn**, and **Pandas** on Kaggle Notebooks.
- Developed a hand gesture recognition pipeline using **OpenCV** and **MediaPipe**, achieving 99% accuracy through careful landmark feature engineering and classifier tuning.
- Conducted data preprocessing, exploratory data analysis (EDA), feature engineering, cross-validation, and performance benchmarking across assigned ML tasks.

CERTIFICATIONS & ACHIEVEMENTS

✓ **Cisco Cybersecurity Essentials** - Cisco Networking Academy

✓ **Python Fundamentals** - Infosys Springboard

✓ **HTML, JavaScript & Bootstrap** - Udemy

✓ **Sawit.ai AI/ML Challenge** - GUVI

- **Open Source Contribution:** Active GitHub contributor with a portfolio of ML and full-stack AI projects at github.com/KDSL13.